

AGUTS

Ahmedabad Gandhinagar Unified Transit System

Milestone 3 — File 1 of 3

Minimal Set of Functional Dependencies

IT-214 DBMS | Winter 2025-26 | DAU | Lab Group 5

Minimal Set of Functional Dependencies

A minimal (canonical) cover $F_{\min}$ has the following properties:

1. Every RHS is a single attribute.
2. No FD in $F_{\min}$ is redundant (removing any FD changes the closure).
3. No attribute on any LHS is redundant (removing it changes the closure).

The 26 relations of AGUTS each have a single-attribute or composite primary key that functionally determines all other attributes. Since every non-key attribute depends directly and fully on the primary key, the minimal FD set contains exactly one FD per relation (or zero for key-only relations).

Summary Table — One FD per Relation

Each row shows the condensed FD. The expanded single-attribute RHS form follows on the next pages.

#	Relation	Candidate Key	Minimal FD Set (condensed)
1	CITY	{city_id}	city_id → city_name, state
2	TRANSIT_OPERATOR	{operator_id}	operator_id → operator_name, service_type, city_id
3	METRO_LINE	{line_id}	line_id → line_name, line_code, operator_id
4	DEPOT	{depot_id}	depot_id → depot_name, area, pincode, operator_id
5	VEHICLE	{vehicle_id}	vehicle_id → registration_no, capacity, manufacture_year, manufacturer_name, manufacturer_info, status, operator_id, depot_id
6	BUS	{vehicle_id}	vehicle_id → fuel_type, bus_type
7	METRO_TRAIN	{vehicle_id}	vehicle_id → train_number, num_coaches
8	MAINTENANCE	{maintenance_id}	maintenance_id → vehicle_id, start_date, end_date, description
9	STOP	{stop_id}	stop_id → stop_name, stop_type, latitude, longitude, city_id, zone_id
10	INTERCHANGE	{interchange_id}	interchange_id → interchange_type, description

11	INTERCHANGE_STOP	{interchange_id, stop_id}	∅ (key-only relation — no non-trivial FDs)
12	FARE_ZONE	{zone_id}	zone_id → zone_name
13	ROUTE	{route_id}	route_id → route_number, route_name, distance_km
14	ROUTE_STOP	{route_id, sequence_no}	{route_id, sequence_no} → stop_id
15	PASSENGER	{passenger_id}	passenger_id → f_name, l_name, email, phone, date_of_birth, concession_type, city_id
16	SMART_CARD	{card_id}	card_id → passenger_id, issue_date, balance, card_status, expiry_date
17	STAFF	{staff_id}	staff_id → full_name, contact, date_of_joining, role, certificate, operator_id, depot_id
18	DRIVER	{staff_id}	staff_id → license_no
19	CONDUCTOR	{staff_id}	staff_id → badge_no
20	MOTORMAN	{staff_id}	staff_id → metro_license_no
21	TRIP	{trip_id}	trip_id → trip_date, etd, eta, trip_status, delay_minutes, route_id, vehicle_id, staff_id
22	TRIP_STOP	{trip_id, stop_order}	{trip_id, stop_order} → stop_id, scheduled_arrival, actual_arrival
23	FARE	{fare_id}	fare_id → amount, concession_type, from_zone_id, to_zone_id
24	PASS	{pass_id}	pass_id → pass_type, service_type, valid_from, valid_to, passenger_id, fare_id
25	TICKET_TRANSACTION	{txn_id}	txn_id → ticket_type, amount, txn_datetime, payment_mode, passenger_id, trip_id, card_id
26	COMPLAINT	{complaint_id}	complaint_id → category, filed_date, description, resolved_date, status, passenger_id, trip_id

Expanded Minimal FDs — Single-Attribute RHS Form

Each functional dependency is expressed with a single attribute on the right-hand side, as required for a strict minimal cover.

R1. CITY

Primary Key: {city_id}

```
city_id → city_name  
city_id → state
```

R2. TRANSIT_OPERATOR

Primary Key: {operator_id}

```
operator_id → operator_name  
operator_id → service_type  
operator_id → city_id
```

R3. METRO_LINE

Primary Key: {line_id}

```
line_id → line_name  
line_id → line_code  
line_id → operator_id
```

R4. DEPOT

Primary Key: {depot_id}

```
depot_id → depot_name  
depot_id → area  
depot_id → pincode  
depot_id → operator_id
```

R5. VEHICLE

Primary Key: {vehicle_id}

```
vehicle_id → registration_no  
vehicle_id → capacity  
vehicle_id → manufacture_year  
vehicle_id → manufacturer_name  
vehicle_id → manufacturer_info  
vehicle_id → status  
vehicle_id → operator_id  
vehicle_id → depot_id
```

R6. BUS

Primary Key: {vehicle_id}

```
vehicle_id → fuel_type  
vehicle_id → bus_type
```

R7. METRO_TRAIN

Primary Key: {vehicle_id}

```
vehicle_id → train_number  
vehicle_id → num_coaches
```

R8. MAINTENANCE

Primary Key: {maintenance_id}

```
maintenance_id → vehicle_id  
maintenance_id → start_date  
maintenance_id → end_date  
maintenance_id → description
```

R9. STOP

Primary Key: {stop_id}

```
stop_id → stop_name  
stop_id → stop_type  
stop_id → latitude  
stop_id → longitude  
stop_id → city_id  
stop_id → zone_id
```

R10. INTERCHANGE

Primary Key: {interchange_id}

```
interchange_id → interchange_type  
interchange_id → description
```

R11. INTERCHANGE_STOP

Primary Key: {interchange_id, stop_id}

No non-trivial FDs (key-only relation)

R12. FARE_ZONE

Primary Key: {zone_id}

```
zone_id → zone_name
```

R13. ROUTE

Primary Key: {route_id}

```
route_id → route_number  
route_id → route_name  
route_id → distance_km
```

R14. ROUTE_STOP

Primary Key: {route_id, sequence_no}

```
{route_id, sequence_no} → stop_id
```

R15. PASSENGER

Primary Key: {passenger_id}

```
passenger_id → f_name  
passenger_id → l_name  
passenger_id → email  
passenger_id → phone  
passenger_id → date_of_birth  
passenger_id → concession_type  
passenger_id → city_id
```

R16. SMART_CARD

Primary Key: {card_id}

```
card_id → passenger_id  
card_id → issue_date  
card_id → balance  
card_id → card_status  
card_id → expiry_date
```

R17. STAFF

Primary Key: {staff_id}

```
staff_id → full_name  
staff_id → contact  
staff_id → date_of_joining  
staff_id → role  
staff_id → certificate  
staff_id → operator_id  
staff_id → depot_id
```

R18. DRIVER

Primary Key: {staff_id}

```
staff_id → license_no
```

R19. CONDUCTOR

Primary Key: {staff_id}

```
staff_id → badge_no
```

R20. MOTORMAN

Primary Key: {staff_id}

```
staff_id → metro_license_no
```

R21. TRIP

Primary Key: {trip_id}

```
trip_id → trip_date  
trip_id → etd  
trip_id → eta  
trip_id → trip_status  
trip_id → delay_minutes  
trip_id → route_id  
trip_id → vehicle_id  
trip_id → staff_id
```

R22. TRIP_STOP

Primary Key: {trip_id, stop_order}

```
{trip_id, stop_order} → stop_id  
{trip_id, stop_order} → scheduled_arrival  
{trip_id, stop_order} → actual_arrival
```

R23. FARE

Primary Key: {fare_id}

```
fare_id → amount  
fare_id → concession_type  
fare_id → from_zone_id  
fare_id → to_zone_id
```

R24. PASS

Primary Key: {pass_id}

```
pass_id → pass_type  
pass_id → service_type  
pass_id → valid_from  
pass_id → valid_to  
pass_id → passenger_id  
pass_id → fare_id
```

R25. TICKET_TRANSACTION

Primary Key: {txn_id}

```
txn_id → ticket_type  
txn_id → amount  
txn_id → txn_datetime  
txn_id → payment_mode  
txn_id → passenger_id  
txn_id → trip_id  
txn_id → card_id
```

R26. COMPLAINT

Primary Key: {complaint_id}

```
complaint_id → category  
complaint_id → filed_date  
complaint_id → description  
complaint_id → resolved_date  
complaint_id → status  
complaint_id → passenger_id  
complaint_id → trip_id
```